VR Headset Ransomware Attack Vulnerability

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***Abstract*—Virtual Reality (VR) Head Mounted Displays (HMD), also called VR headsets, are breaching barriers that previously held back widespread adoption and usage. The escalation in usage prompts the idea that standalone HMD units can become malware targets. This work explores the applicability of Android ransomware to the Oculus Quest 2's attack surface. To test the environment ten families of Android ransomware samples were applied along with a proof-of-concept ransomware sample that was developed for this research. All experiments are compared through malware execution on Oculus Quest-2. Observed ransomware sample behavior is compared to the expected behavior of each ransomware sample family, as well as to the derived ransomware definition. Ransomware sample success is evaluated based on expected behavior and the ability of the samples to execute definitional ransomware traits. The research indicates that the Oculus Quest 2's attack surface contains the necessary aspects for the successful execution of ransomware.**

Keywords— Malware, Oculus Quest 2, Virtual Reality, Ransomware.

INTRODUCTION

It is predicted that the projected revenue of the Augmented Reality (AR) and Virtual Reality (VR) environment in 2024 will be approximately 38.6 billion in 2024 and will have a projected market value of 58.1 billion by 2028 [1]. They also state that user penetration is projected to be 52.8% in 2024 and around 55.9% in 2028. Continuous improvements in VR technologies not only encourage market growth but enable the utilization of this equipment in sensitive sectors like rehabilitation and medical areas [2]. Gaming technologies, like Nintendo Wii and Microsoft Kinect, are capitalizing on this opportunity by integrating headset use into patient rehabilitation protocols and demonstrating positive initial outcomes [3], [4], [5], [6]. According to Statista, the best-selling VR HMDs are the Oculus headsets and the Quest 2 (OQ2), which are at the forefront for Steam Gamers [1]. The OQ2 runs a modified Android 10 Operating System (OS) [7]

According to StatCounter, in 2023, 41.64% of all devices worldwide used Android as their OS [8]. Another report states that the rapid growth of the Android market attracts attackers seeking unauthorized access to Android devices and data using malware applications [9]. The authors go on to indicate that compared to iOS, Android is at a higher risk of experiencing security breaches and malware attacks. In support of this claim, recent studies report a significant rise in ransomware attacks [10].

By utilizing malicious code, ransomware infiltrates victims' device OS and blocks their data unless they pay a ransom [11]. Ransomware developers have created various sophisticated methods to cause financial harm to their victims [12]. As a result, ransomware has become one of the most threatening types of attacks that target both organizations and individuals, with financial losses reaching billions of dollars, in addition to one million Android users in a single month have been affected by ransomware, causing significant harm [13]. Additionally, the recent success of ransomware has led to the emergence of new families of this type of malware, such as Wannacry, Jigsaw, Satana, Petya, Shade, Cryptomix, CTB-Locker, and many others [14].

Over the last couple of years, ransomware has played a major role in the marked decline of cyber insurers' ability to evaluate risks effectively [15]. Overall, the loss ratio for U.S. cyber insurance increased from 44.6% in 2019 to 66.9% in 2020, with ransomware being responsible for three-quarters of the claims according to credit rating agency AM Best [16]. As stated by the coalition's cyber insurance claim report [17], 41% out of 87% of reported attacks result from incidents involving ransomware attacks. Additionally, ransomware can take two forms: locker-ransomware and crypto-ransomware [18]. Specifically, Locker ransomware restricts the victim's access to their device until a ransom is paid, while crypto ransomware encrypts personal files to make them inaccessible to the owner.

Coupling the increase in AR and VR revenue with rising user penetration, integrated functionality in healthcare markets, and escalating ransomware implementations prompts the hypothesis that Android ransomware is executable on the OQ2. This hypothesis raises the following research questions:

- Can malware be successfully installed on existing Oculus Quest 2 devices?
- Can existing Android ransomware code be executed on an Oculus Quest 2 device?
- How effective is the ransomware on the device?

The contribution of this research is two-fold. First, this research provides data supporting the feasibility of ransomware execution on an OQ2 standalone headset. Secondly, it provides the foundation for future research into the viability of

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ransomware on standalone virtual reality devices. This research also introduces a Simple Ransomware Sample (SRS) as a Proof-of-Concept, a minimum viable ransomware for examining ransomware on the attack surfaces of Android devices. The balance of the paper is structured in the following manner. Section 2 examines current research in the field of VR security. Section 3 presents the methods and tools used to conduct the research. Section 4 covers the results and analysis. Section 5 concludes and discusses future work.

LITERATURE REVIEW

With any technology, an attack surface exists, i.e., the total area of exposed functionality and system resources that present vulnerabilities that malicious actors may exploit [19]. While compromising attack surfaces is often an interest of practitioners and academicians [20], [21], [22], testing attack surfaces provides insight into the mitigation or even prevention of attacks [23], [24].

Hence, this section examines current research in the field of VR HMD security [25]. The range of security research includes exploring authorization methods, such as virtual keyboards and other novel methods of password input security [26]. Current security research introducing the Man-in-the-Room (MitR) attack, exploiting vulnerabilities in social networking apps for VR [27]. For VR system forensics [28], emphasizing data recovery potential and the need for further AR and VR forensic research.

Yarramreddy's research team [28] conducted a comprehensive exploration into the forensics of immersive Virtual Reality (VR) systems and their social applications, including Facebook Spaces, Rec Room, Altspace VR, and Bigscreen. They focused on widely applied consumer VR systems, namely the Oculus Rift and the HTC Vive, examining the efficacy of reconstructing evidence from the systems themselves and network traffic. The author's findings indicated the potential recovery of significant forensically relevant data such as system details, events, user profile pictures, and usernames. Lastly, the work underscored the need for further research in AR and VR forensics.

In their research [27], they applied classic exploitation methods against modern technology to acquire equally novel outcomes. The unique attributes of the VR landscape prepare the groundwork for their main account of a new attack, the Man-in-the-Room (MitR). From a vulnerable social networking application, this attack originated, and led to the adaptation of worming and botnet capabilities for VR, capable of significantly affecting millions of individuals. Their research enhanced the state-of-the-art socio-technical and VR security research in VR. It shares several attacking and analytical tools, like evaluation dataset, exploits, and vulnerability signatures with the scientific communities to guarantee the development of secure VR software. Results indicate the prevention and detection of VR vulnerabilities.

Ling and colleagues [29], introduced novel motion sensor-based and computer vision-based side-channel attacks in a virtual environment to drive keystrokes. In the motion sensor-based attack, they exploit sensors in particular the gyroscope, magnetometer, and accelerometer embedded within the pointing device to extract its orientation angles. Using the orientation angles of the pointing device, authors inferred the empirical rotation angles and derived the clicked keys by measuring the similarity between the reference ones among keys and the empirical rotation angles. On the other hand, a stereo camera captures a user's actions as they manipulate a pointing device and enter passwords in the context of the computer vision-based attack. Several computer vision approaches are used to identify the touching frames when the user touches the touchpad to input keys on the virtual keyboard within the virtual environment. Subsequently, they analyzed the headset's position to determine the orientation of the virtual pointing ray during each touching frame.

Odeleye's research group [30] showed that modifications in framerate can cause VR sickness. The authors then set up and designed tests for different techniques of framerate-degrading attacks. The original form of attack involved a malware application that released and used Graphics Processing Unit (GPU) resources to manipulate the framerate on-device. The HMD crashed when the GPU was exposed to a high computational load. Prior to the HMD crash, symptoms like screen tears and dropped frames occurred. Dropped screen tears and frames can cause visual disorientation for a user.

A second aproach that Odeleys' group studied involved building a local Denial of Service (DoS) attack via the HMD network interface. During the DoS attack, the target HMD experienced a flood of Internet Control Message Protocol (ICMP) pings. Framerate fluctuations occurred because of the DoS attack's overwhelming system processing speed. Both attack techniques caused framerate problems, which can cause VR sickness in a user. The team also produced a mitigation method to framerate attacks through using machine learning-based detection. Lastly, the researchers suggested employing this detection method to notify users of an ongoing attack and protect them, which gives them a chance to mitigate the symptoms of VR sickness and remove their HMD.

Mortezapoor and Vasylevska [31], discussed the difficulties associated with integrating VR systems and robots using Robot Operating System (ROS). Also, they expressed concern about the potential security risks associated with sharing data. Ensuring user safety is a top priority when using robots for simulations in VR, which necessitates collecting and sharing redundant data. Therefore, the community should carefully investigate the trade-offs between privacy/security and system safety/flexibility. Previous research covers emerging VR attack points. While attacks primarily focus on properties unique to VR HMDs, minimal research investigates the application and effectiveness of existing Android ransomware to OQ2.

METHODOLOGY

VR platforms are split into standalone and tethered models [32]. Tethered models do not themselves run operating systems and are merely input/output devices requiring a host computer to operate [33]. Examples of tethered models include the Valve Index and the HTC Vive [32], [33]. Standalone systems have dedicated storage, run their own OSs, and do not require another computer to operate [34]. Because the tethered models

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do not act as host devices, malware targeting them would be targeting their host systems.

Hence, this work expands the exploration of ransomware on standalone OQ2 VR HMDs initiated by [35] to investigate the viability of Android ransomware executing on the OQ2's attack surface. To study ransomware viability on an OQ2 device, a controlled experiment was conducted based on [36]. TABLE I. provides software and system versioning information for the experiment environment. For research purposes, all other headsets and ransomware are considered out of scope.

TABLE I. VERSIONING INFORMATION

System	Information
Android SDK	Android 10.0 SDK, API version 29, Revision 5
OQ2 Software Update	System Version 22310100587300000
OQ2 Version	37.0.0.147.109.346379382
OQ2 Runtime Version	37.0.0.147.110.346379395
OQ 2 OS Version	user-22310100587300000
Computer	Dell Inspiron 3847 500GB, Intel Core I5, 3.1GHz, Windows 7
Cellular Phone	Samsung Galaxy A30s MSM-A307FNMZSRZ8N10NZNMZ, Ver.11
Oculus App	iOS platform, version 152.0

The controlled experiment is divided into two stages: the pretest configuration, illustrated in Fig. 1, followed by the ransomware test stage.

A. OQ2 Pretest Configuration

The OQ2 Pretest Configuration began by setting up an Oculus/Meta account on the computer, creating an unmerged Oculus Developer Account, and completing verification steps. Then, an organization was established within the account after setting up 2-Factor Authentication (2FA). A test user was generated within this organization on the computer by creating an account linked to a dummy Meta account. Afterward, the Oculus phone app was signed into using the test user's credentials, followed by pairing a new headset and completing necessary setup steps such as language selection and Wi-Fi connection. Finally, the Oculus Quest 2 was connected to the SideQuest app by enabling Developer Mode on the device, connecting it to the computer via USB cable, and managing apps and games through SideQuest.

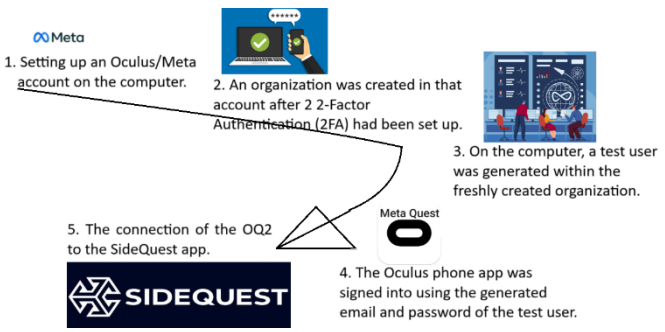


Fig. 1. The OQ2 Pretest Configuration Steps

B. Ransomware Test Process

The Ransomware Test Process involves a systematic series of steps to assess the vulnerability of the Oculus Quest 2 (OQ2) to ransomware. To maintain a pristine testing environment, the OQ2 was reset to its factory default state before each round of testing. Following this, the OQ2 underwent its initial setup, including a Wi-Fi network connection, and the Oculus app on a mobile phone was used to pair with the device. Next, the developer mode was enabled through the app to facilitate the sideloading of ransomware samples. However, to grant necessary permissions, USB file access from a computer and debug access were allowed on the OQ2. The setup for voice commands, crucial for capturing sample behaviors, was completed. The actual ransomware samples were installed on the OQ2 using the SideQuest application on the computer.

Screenshots were captured at various points in the process, from the initial view to the permissions request view and any other relevant screens. Each ransomware sample was activated, and all available views were documented. To achieve this, the Apps menu on the OQ2 was accessed, followed by selecting "Unknown Sources," with necessary permissions granted for testing purposes. Following these assessments, screenshots were synchronized in the Gallery on the OQ2, and the phone app was used to manage and move these screenshots to a separate file. Finally, these steps were meticulously repeated for each ransomware sample to ensure a comprehensive evaluation of their behavior and impact on the OQ2, ultimately contributing to a comprehensive understanding of its security vulnerabilities. The ten (10) specific types of Android ransomware are presented in TABLE II.

TABLE II. USED RANSOMWARE TYPES

Ransomware	Description
Charger	Masquerades as a phone charging application. The application harvests user data, and some variants also request a bitcoin ransom [37]
Jisut	Fake lock screens and demands for ransoms [37]
Koler	Does not encrypt or manipulate any files but does create a persistent window that will not close and persists between device reboots [38]
Lockerpin	Creates and/or sets the PIN to unlock the infected Android device [37]
Pletor	Encrypts files, demands a financial ransom and locks the device [39]
RansomBO	Encrypts files based on commands from a command and control [40]
Simplocker	Encrypts files on the infected device, and demands a financial ransom [37]
Svpeng	Screen locker ransomware demands a ransom under the guise of threatening to report the user to authorities. [39]
WannaLocker	Immediately activates and encrypts files and asks for ransom to decrypt files [41]
SARA	This attack locks the device, demanding payment for the unlock key [42]

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Simple Ransomware Sample (SRS)

The Last test sample is SRS [35], a proof-of-concept ransomware developed for this research. SRS was developed to examine the design principles of ransomware in depth since it fulfills the twofold definition of Preventing entry to system resources and then demanding a ransom from the user. The SRS was created as a standard Android application for portability, without using any part of the Oculus SDK. The Kotlin language was used to write the SRS code, along with the Android 10 Google Pixel VM for development and testing. This Kotlin code was compiled into Java Virtual Machine (JVM) bytecode. For encryption/ decryption, researchers used Java's Crypto and Security libraries, as well as cipher generation and cipher-related components from this library for encrypting files was employed.

SRS on load utilizes a combination of functions known as checkPerms() and genList() to generate a list of authorizations the application presently has. If the application lacks certain necessary permissions to function, then by using the requestPerms() it prompts the user to grant those permissions. The code was compiled using Android Studio "APK" and sideloaded by Android Studio's interface.

The idea is to deceptively introduce SRS to users as a password-based file encryption program. The victim inputs a password and has the option to decrypt or encrypt data. SRS was activated through the 'Encrypt' button, after obtaining permissions to read and write data it proceeded to encrypt all available data by utilizing the findFiles() function and the encrypt() function. However, the password provided by the user is not used for encryption; rather, the password provided by the attacker is used. Accordingly, when a user tries to decrypt their data, SRS displays a message instructing the victim to contact the attacker.

DISCUSSION

After applying the Android ransomware samples to an OQ2, the results showed that seven out of ten samples were installed, applied, and successfully affected the device. The samples that worked are Charger, jisut, koler, lockerpin, sara, somplocker, and wannalocker as shown in Fig. 2-5. The ransomware impact was split into two types of impact. The first type of impact affects data. Data was encrypted by Jisut and WannaLocker. The second type of impact affects the device itself; device features were disabled by Charger, Koler, Lockerpin, Simplocker, and SARA. For example, after running the Lockerpin, the voice command and microphone were disabled, which prevents taking screenshots, as this is supposed to be accomplished using voice commands. Hence, a picture was acquired using a mobile phone, as seen in Fig. 3.

As for Svpeng and Pletor, the device starts disconnecting and restarting. As a result, the ransomware is not able to display its message, and users can't tell whether the device was a victim of a ransomware attack or just malfunctioning. Other possible reasons for ransomware failure include corrupt installation files and operating system compatibility issues. The ransomware results are presented in TABLE III.

TABLE III. RANSOMWARE ACTIVATION RESULTS

Ransomware	Results: Symptoms
Charger	Successful: Screen shaking
Jisut	Successful: Fake lock appeared and asked for ransom.
Koler	Successful: An uncloseable window was created and persisted between device reboots.
Lockerpin	Successful: the device was locked, asked for a PIN.
Pletor	Unsuccessful: Nothing happens for the device.
RansomBO	Unsuccessful: Nothing happens for the device.
Simplocker	Successful: blocked the device, payment required.
Svpeng	Unsuccessful: Nothing happens to the device.
WannaLocker	Successful: The device's files were encrypted.
SARA	Successful: locked the device, payment required.



Fig. 2. Somplocker

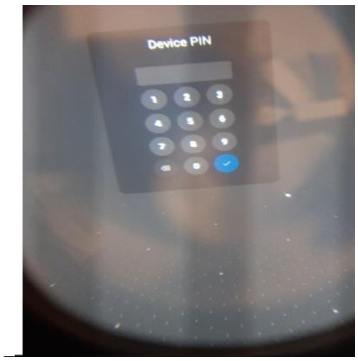


Fig. 3. Lockerpin



Fig. 4. Sara

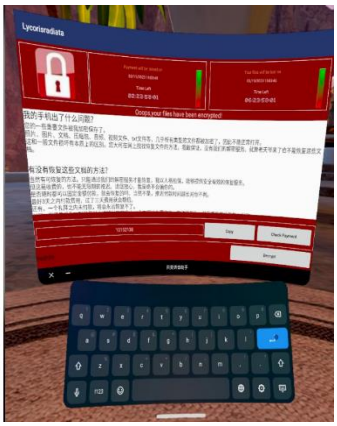


Fig. 5. Wannalocker

In addition to the ten samples tested, SRS successfully encrypted, relocated and renamed all test files. As expected, it effectively blocked them from access and ransomed the blocked resources. The attack surface vectors were permissions to a user's device storage to read and write, and those vectors were present. SRS worked as designed and met the criteria of ransomware's main definition. The process of generating a cipher object and utilizing SecureRandom() is shown in Fig. 6.

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```
// Key size 16 bytes!
val sks = SecretKeySpec(key.toByteArray(), "AES")
val cipher = Cipher.getInstance("AES/CBC/PKCS5Padding")
//gen ivz
val rnd = SecureRandom()
val iv = ByteArray(16)
rnd.nextBytes(iv)
val spec = IvParameterSpec(iv)
output.write(iv)
Log.e("DEBUG", "iv for $address is $iv")
```

Fig. 6. SecureRandom() used to generate 16 byte IVs in the cipher object creation.

For encryption purposes, The attacker's password was used as shown in Fig. 7.

```
class MainActivity : AppCompatActivity() {
    private lateinit var permList: Array<String>
    private lateinit var list: List<item>
    private lateinit var recycler_view: RecyclerView
    private val realKey: String = "trickypassword12" //scammer password
```

Fig. 7. The main view class initializes the attacker's password.

As indicated in Fig. 8, the encrypt function is exclusively given the password of the attacker every time it's called. The encrypted data is inaccessible to the user and can only be decrypted using the attacker's password. Whenever the user tries to decrypt the data with an incorrect password, the attacker's contact information is revealed, see the Decrypt button code in Fig. 9.

```
encryptButton.setOnClickListener {
//                val key: String = passwordTxt.text.toString()
//deactivated for attack
                Log.e("DEBUG", "got pass $realKey")
                val files: ArrayList<File> = findFiles(store)
                Log.e("DEBUG", "got files")
                if (files.isNotEmpty()) {
                    Log.e("DEBUG", "files not empty")
                    for (file in files) {
                        //we send them to encrypt 1 x 1, passing the
name of the file and its location
                        Log.e("DEBUG", "got a $file")
                        encrypt(realKey, file.path, file.name)
                    }
                }
                Log.e("DEBUG", "done")
            }
            Log.e("DEBUG", "set encrypt button")
```

Fig. 8. The Encrypt button uses the attacker's password, not the user's.

```
decryptButton.setOnClickListener{
    val key: String = passwordTxt.text.toString()
    if (key.isNotEmpty()) {
        Log.e("DEBUG", "got pass $key")
        if (key == realKey) {
            val files: ArrayList<File> = findFiles(store)
            for (file in files) {
                //we send them to decrypt 1 at a time, passing the name
of the file and its location, but first
                // we check if it is encrypted, (that in its name it
has the word "encrypt_" to prevent double encryption
                val check: Int = file.name.indexOf("encrypt_")
                if (check != -1) {
                    Log.e("DEBUG", "undoing a $file")
                    decrypt(key, file.path, file.name)
                }
            }
        } else {
            Toast.makeText(this,"Contact scammer@mail.com for real
password", Toast.LENGTH_SHORT).show()
        }
```

Fig. 9. Decrypt button on-click function. The user's input key is compared to the attacker's key.

Conclusion

Based on the results of this study, it is evident that VR HMDs, particularly OQ2, are not immune to ransomware attacks. The experiments with Android ransomware samples showed that almost all the ransomware samples tested, specifically Charger, jisut, koler, lockerpin, sara, somplocker, wannalocker, along with SRS were able to install and execute on the OQ2 successfully. The investigation also exposed the efficiency of the ransomware that was reflected by encrypting the device data in some scenarios and disabling the device functionality in other scenarios. However, the study also revealed limitations in the device's attack surface. Some ransomware samples could not fully execute due to device resources being locked or the device disconnecting and restarting. Altogether, this research highlights the importance of improving the security measures for VR devices to prevent potential malware attacks.

Overall, the findings of this research support the hypothesis that Android ransomware is executable on the OQ2's attack surface. The results indicate that OQ2 is potentially vulnerable to ransomware attacks and that the need for appropriate security measures should not be overlooked. As the popularity and usage of VR devices continue to grow, it is crucial to integrate effective security measures to ensure the safety and protection of users.

Future Work

This study focused specifically on the OQ2 and Android ransomware samples. Future work will involve testing a wider range of VR, Extended Reality (XR), and AR HMDs like Hololens2 or HTC VIVE headsets against expanded malware samples to gain a better understanding of the overall attack surface and potential vulnerabilities. Future explorations will also involve the development of more targeted technical attacks and security measures that specifically address the unique characteristics and potential attack vectors of VR, AR, and XR HMDs. Additionally, specific industries and use cases will be targeted to investigate the development of mitigation strategies.

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